Prepared By	08-Aug-2012	Mr.Chatchawan S.	RNC RENEWCOILS ENGINEER & SUPPLY CO.,LTD.	
Checked By	08-Aug-2012	Mr.Chatchawan S.		
Approved By	08-Aug-2012	Mr.Amnauy W.		
Certified By	08-Aug-2012	Mr.Amnauy W.	Renew coils Engineering & Supply Co., Ltd.	
Rev.	Date	Checked	Approved	Description
0	08-Aug-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-015-2012

Addendum

WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-002

WELDING PROCEDURE SPECIFICATION (WPS)

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

WELDING PROCEDURE SPECIFICATION (WPS) Yes ■
PREQUALIFIED X QUALIFIED BY TESTING ____
or PROCEDURE QUALIFICATION RECORD (PQR) ☐

Identification # RNC-WPS-002
Revision 0 Date 8-Aug-12 By Mr. Chatchawan s.
Authorized by Mr. Amnuy V. Date 6-Aug-12

Company Name: Renew coils and Engineering & Supply co.,ltd.
Welding Process (es): SMAW
Supporting PQR No.(s): RNC-PQR-002

Type-Manual ■
Machine ☐
Semi-Automatic ☐
Automatic ☐

JOINT DESIGN USED
Type : Butt Weld
Single ■ Double Weld ☐
Backing : Yes ☐ No ■
Backing Material : NONE
Root Opening 3 Root Face Dimension 3
Groove Angle : 60$^{0\pm5}$ Radius (J-U) NONE
Back Gouging : Yes ■ No ☐ Method

POSITION
Position of Groove : 2G Fillet : None
Vertical Progression : Up ■ Down ☐

ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW) Short-Circuiting ☐ Globular ☐ Spray ☐
Current : AC ■ DCEP ☐ DCEN ☐ Pulsed ☐
Other
Tungsten Electrode (GTAW)
Size : NONE
Type : NONE

BASE METALS
material Spec. SS400 / A36 or Strength Equivalence
Type or (Grade) Group 1 Carbon
Thickness : Groove 3.0 mm - Unlimited Fillet N/A
Diameter (Pipe) -

TECHNIQUE
Stringer or Weave Bead : Stringer and Weave Bead
Multi-pass or Single Pass (per side) Multi-pass
Number of Electrodes
Electrode Spacing Longitudinal N/A
Lateral N/A
Angle N/A

FILLER METALS
AWS Specification A 5.1
AWS Classification E7016
Brand Name KOBELCO LB 52

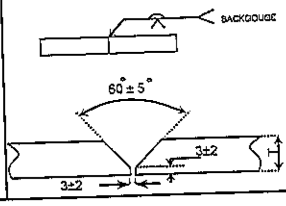
SHIELDING
Flux N/A Gas N/A
Composition
Electrode-Flux (Class) N/A Flow Rate
Gas Cup Size

Contact Tube to Work Distance
Peening N/A
Interpass Cleaning : Grinding or Wire Brush

PREHEAT
Preheat Temp., Min 30 C°
Interpass Temp., Min 100C° Max 250 C°

POSTWELD HEAT TREATMENT
Temp. N/A
Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals		Current		Volts	Travel Speed cm/min	Joint Details
		Class	Diam.	Type & Polarity	(Amps) or Wire Feed Speed			
Root pass	SMAW	E7016	3.2	AC/EP	90-130	20-25	6-12	
Hot pass	SMAW	E7016	3.2	AC/EP	90-130	20-25	6-12	
Filling pass	SMAW	E7016	4	AC/EP	130-180	25-28	10-18	
cap	SMAW	E7016	3.2	AC/EP	90-130	20-25	6-12	

PREPARED BY : [signature]
DATE : 18/08/12

REVIEWED BY : [signature]
DATE : 18/08/12

APPROVED BY : [signature]
DATE : 20-8-12

WELDING PROCEDURE SPECIFICATION (WPS)

RNC
RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

WELDING PROCEDURE SPECIFICATION (WPS) Yes [x]
PREQUALIFIED X QUALIFIED BY TESTING ___
or PROCEDURE QUALIFICATION RECORD (PQR) ___ [ ]

Identification # RNC-WPS-002
Revision 0 Date 8-Aug-12 By Mr. Chatchawan s.
Authorized by Mr. Amnuy V. Date 6-Aug-12
Type-Manual [x] Semi-Automatic [ ]
Machine [ ] Automatic [ ]

Company Name: Renew coils and Engineering & Supply co.,ltd.
Welding Process (es): SMAW
Supporting PQR No.(s): RNC-PQR-002

JOINT DESIGN USED
Type: Butt Weld
Single [x] Double Weld [ ]
Backing: Yes [ ] No [x]
Backing Material: NONE
Root Opening 3 Root Face Dimension 3
Groove Angle: $60^{0\pm5}$ Radius (J-U) NONE
Back Gouging: Yes [x] No [ ] Method

BASE METALS
material Spec. SS400 / A36 or Strength Equivalence
Type or Grade: Group 1 Carbon
Thickness: Groove 3.0 mm - Unlimited Fillet N/A
Diameter (Pipe) -

FILLER METALS
AWS Specification: A 5.1
AWS Classification: E7016
Brand Name: KOBELCO LB 52

SHIELDING
Flux N/A Gas N/A
Composition
Electrode-Flux (Class) N/A Flow Rate
Gas Cup Size

PREHEAT
Preheat Temp., Min 30 C°
Interpass Temp., Min 100C° Max 250 C°

POSITION
Position of Groove: 2G Fillet: None
Vertical Progression: Up [x] Down [ ]

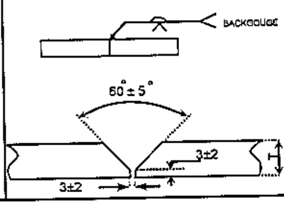
ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW) Short-Circuiting [ ]
Globular [ ] Spray [ ]
Current: AC [x] DCEP [ ] DCEN [ ] Pulsed [ ]
Other
Tungsten Electrode (GTAW)
Size: NONE
Type: NONE

TECHNIQUE
Stringer or Weave Bead: Stringer and Weave Bead
Multi-pass or Single Pass (per side): Multi-pass
Number of Electrodes
Electrode Spacing: Longitudinal N/A
Lateral N/A
Angle N/A

Contact Tube to Work Distance
Peening N/A
Interpass Cleaning: Grinding or Wire Brush

POSTWELD HEAT TREATMENT
Temp. N/A
Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals Class	Filler Metals Diam.	Current Type & Polarity	Amps or Wire Feed Speed	Volts	Travel Speed cm/min	Joint Details
Root pass	SMAW	E7016	3.2	AC/EP	90-130	20-25	6-12	
Hot pass	SMAW	E7016	3.2	AC/EP	90-130	20-25	6-12	
Filling pass	SMAW	E7016	4	AC/EP	130-180	25-28	10-18	
cap	SMAW	E7016	3.2	AC/EP	90-130	20-25	6-12	

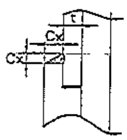
PREPARED BY: DATE: 18/08/12

REVIEWED BY: DATE: 18/08/12

APPROVED BY: DATE: 20-8-12

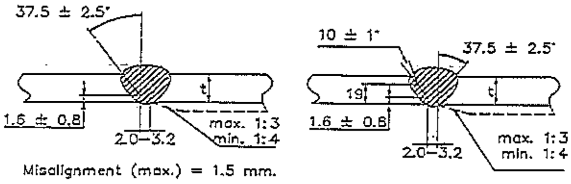
RNC-SMAW-001

ATTACHED FIGURE JOINT SKETCH



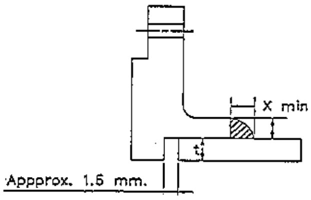
Cx (min.)=1.25t but not less than 3.2 mm.

JOINT No. A1



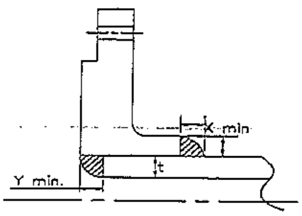
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



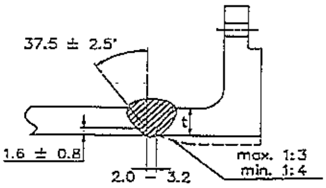
X min. = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



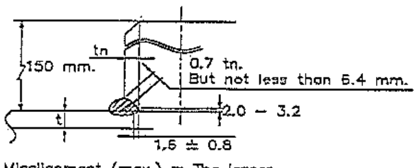
X min. = The lesser of 1.4t or thickness of the hub.
Y min .= The lesser of t or 6.4 mm.

JOINT No. A3



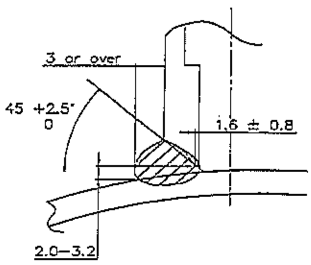
Misalignment (max.) = 1.5 mm.

JOINT No. B3

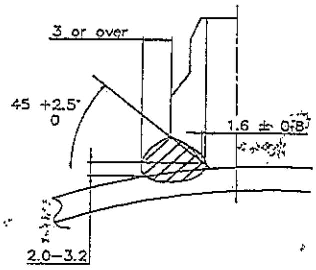


Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

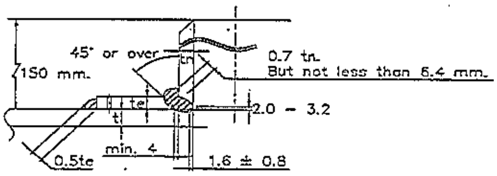
JOINT No. B4



JOINT No. C1

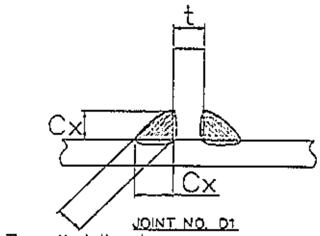


JOINT No. C2

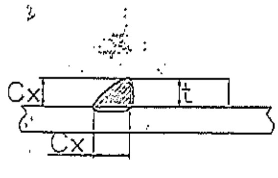


Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

JOINT No. B5



JOINT NO. D1



JOINT NO. D2

Note : Cx = Size of weld as specified
Theoretical throat = 0.707 x Cx

PAE TECHNICAL SERVICE COMPANY LIMITED
69 Soi On-nuch 64, Srinakarin Road, Suanluang, Suanluang, Bangkok 10250
Tel. : (662) 322-0222 (Auto 20 lines) Fax. : (662) 721-2577

WELDING PROCEDURE SPECIFICATION (WPS) Yes ☐
PREQUALIFIED ______ QUALIFIED BY TESTING ______
or PROCEDURE QUALIFICATION RECORD (PQR) Yes ☒
(ACCORDANCE WITH THE REQUIREMENT OF SEC'T 4 AWS D1.1 2010)

Identification # RNC-WPS-002
Revision 0 Date ______ By Mr.Chatchawan S.
Autorized by Mr.Amnuy V. Date ______
Type ☑ Manual ☐ Semi-Automatic ☐ Machine ☐ Automatic

Company Name: Renew Coils Engineering & Supply Co.,Ltd.
Welding Process(es): SHIELD METAL ARC WELDING
Supporting PQR No.(s): RNC-PQR-002

JOINT DESIGN USED
Type : Single V-Groove Butt Weld
☑ Single ☐ Double weld
Backing : ☐ Yes ☑ No
Backing Material : None
Root Opening : 3.0 mm. Root Face Dimension : 1.0 mm.
Groove Angle : $60^{\circ \pm 5}$ Radius (J-U) -
Back Gouging ☑ Yes ☐ No
Method : Grinding

POSITION
Position of Groove 2G Fillet F,H
Vertical Progression ☐ Up ☐ Down

ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW) ☐ Short-Circuiting ☐ Globular ☐ Spray/Pulse
Current ☑ AC ☐ DCEP ☐ DCEN ☐ Pulsed
Other -
Tungsten Electrode (GTAW)
Size None
Type None

BASE METALS
Material Spec. ASTM,JIS G3101
Type or Grade SS 400
Thickness : Groove 25 mm. Fillet : N/A
Diameter (Pipe) -

TECHNIQUE
Stringer or Weave Bead : Stringer
Multi-pass or Single Pass (per side) Multipass
Number of Electrodes Single
Electrode Spacing Longitudinal N/A
Lateral N/A
Angle N/A

FILLER METALS
AWS Specification A 5.1
AWS Classification LB-52/E7016

SHIELDING
Flux - Gas -
Composition -
Electrode-Flux(Class) - Flow Rate -
Gas Cup Size -

Contact End to Work Distance Arc stable
Peening None
Interpass Cleaning : Slags remove by Grinding / Wire Brush

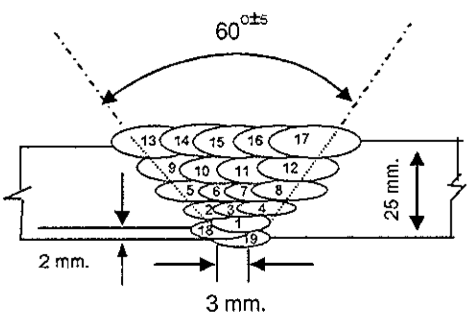
PREHEAT
Preheat Temp.: Ambient
Interpass Temp., Min 30^{0}C Max 246° C

POSTWELD HEAT TREATMENT
Temperature N/A
Time N/A

WELDING PROCEDURE

Pass or Weld Layer(s)	Process	Filler Metals Class	Filler Metals Dia. (mm.)	Current Type & Polarity	Current Amperage or Wire speed	Volts	Travel Speed (cm / min)	Heat Input (kj/cm)
1	SMAW	E7016	3.2	DCEP	110	22	5.41	26.84
2	SMAW	E7016	3.2	DCEP	141	25	6.98	30.30
3	SMAW	E7016	3.2	DCEP	165	26	5.56	46.30
4	SMAW	E7016	3.2	DCEP	168	26	7.99	32.81
5	SMAW	E7016	3.2	DCEP	166	26	9.95	26.03
6	SMAW	E7016	3.2	DCEP	164	26	10.27	24.92
7	SMAW	E7016	3.2	DCEP	165	26	9.86	26.11
8	SMAW	E7016	3.2	DCEP	166	26	10.00	25.90
9	SMAW	E7016	3.2	DCEP	164	26	14.29	17.91
10	SMAW	E7016	3.2	DCEP	168	26	10.61	24.71
11	SMAW	E7016	3.2	DCEP	165	26	14.53	17.72
12	SMAW	E7016	3.2	DCEP	165	26	10.58	24.33
13	SMAW	E7016	3.2	DCEP	166	26	13.95	18.57
14	SMAW	E7016	3.2	DCEP	165	26	16.51	15.59
15	SMAW	E7016	3.2	DCEP	167	26	14.83	17.57
16	SMAW	E7016	3.2	DCEP	165	26	15.35	16.66
Back Weld	SMAW	E7016	3.2	DCEP	164	26	15.63	16.37
18	SMAW	E7016	3.2	DCEP	165	26	11.01	23.38
19	SMAW	E7016	3.2	DCEP	164	26	6.87	37.24

Joint Details



PAE TECHNICAL SERVICE COMPANY LIMITED
69 Soi On-nuch 64, Srinakarin Road, Suanluang, Suanluang, Bangkok 10250
Tel. : (662) 322-0222 (Auto 20 lines) Fax. : (662) 721-2577

PQR No. : RNC-PQR-002

PROCEDURE QUALIFICATION RECORD (PQR)

NDT and Mechanical Test Result

TENSILE TEST (Report No : TS-12-08-080)

Specimen No.	Width mm.	Thickness mm.	Area	Ultimate Load (KN)	Ultimate Strength ($N/mm.^2$)	Character of Failure and Location
3179/12 TR1	20.31	24.93	506.328	219.276	433.071	OUT OF WELD
3179/12 TR2	20.07	24.79	497.535	216.727	435.602	OUT OF WELD

GUIDE BEND TEST (Report No : BD-12-08-019)

Specimen No	Type of Bend	Test Result	Location of Indication
3179/12 BS1	Transverse Side Bend	Open Discontinuity 0.58 mm	WELD ZONE
3179/12 BS2	Transverse Side Bend	No Open Discontinuity	N/A
3179/12 BS3	Transverse Side Bend	No Open Discontinuity	N/A
3179/12 BS4	Transverse Side Bend	No Open Discontinuity	N/A

Visual Inspection, PQR Report No: PQR-009

Appearance: Acceptable
Undercut/Slags: No Indication
Surface Porosity: No Indication
Test Date: 28 Jul 12
Witness By: Mr.Wichit K.
Organization: PAE Technical Service Co.,Ltd.

Radiographic Examination

RT Report No : RT-PQR-014 Result : Accepted
Organization : PAE Technical Service Co.,Ltd.

Other Tests

1) **Vicker Hardness Test**
Report No: HN-12-08-032

2) **Macrostructure Examination**
Report No: MA-12-08-023

3) **Fillet Weld Test**
Report No: **N.A.**

4) **Impact Test**
Report No: IM-12-08-085

Welder's Name : Mr.Boontham J. **Welder No :** -
Mechanical Test Conducted By : LPN Plate Mill Public Company Limited
Test Conducted By : Mr.Keerati Sa-ngoen **Report No :**

We, the undersigned,certify the statements in this record are correct and that the test welds were prepared,welded, and Tested in conformance with the requirements of Clause Section 4 of AWS D1.1 Strutural Welding Code-Steel.,

Manufacturer by State Construction Co.,Ltd.	Contractor by PAE Technical Service Co.,Ltd.
Signature:	Signature:
Name: Mr. Chatchawan R.	Name: Mr. Chaloemkiet R.
Date : 17/08/12	Date : 17-Aug-2012

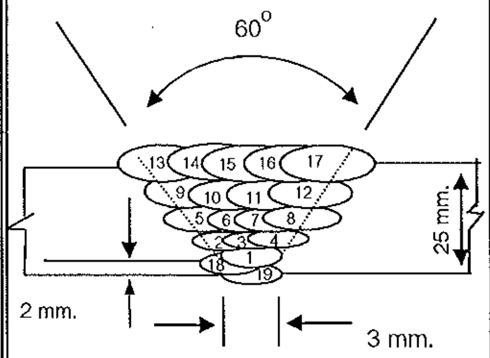
PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINERING CONSULTANT
69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Client :	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-009
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-002	Date : 6-Aug-12

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.BOONTHAM J.	SMAW	2G	AWS D1.1	JIS G3101 Gr.SS400	25x290x380	A 5.1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	SMAW	E 7016	3.2	DCEP	110	22	5.41	30	26.84	-	-	-	-
2	SMAW	E 7016	3.2	DCEP	141	25	6.98	97	30.30	-	-	-	-
3	SMAW	E 7016	3.2	DCEP	165	26	5.56	148	46.30	-	-	-	-
4	SMAW	E 7016	3.2	DCEP	168	26	7.99	212	32.81	-	-	-	-
5	SMAW	E 7016	3.2	DCEP	166	26	9.95	213	26.03	-	-	-	-
6	SMAW	E 7016	3.2	DCEP	164	26	10.27	235	24.92	-	-	-	-
7	SMAW	E 7016	3.2	DCEP	165	26	9.86	234	26.11	-	-	-	-
8	SMAW	E 7016	3.2	DCEP	166	26	10.00	241	25.90	-	-	-	-
9	SMAW	E 7016	3.2	DCEP	164	26	14.29	212	17.91	-	-	-	-
10	SMAW	E 7016	3.2	DCEP	168	26	10.61	213	24.71	-	-	-	-

PAE Inspection By	Witness By	Approved By
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Wichit K.	NAME : Mr Chatchawan S.	NAME : Mr Chatchawan S.
DATE : 7-Aug-2012	DATE : 7/08/12	DATE : 7/08/12

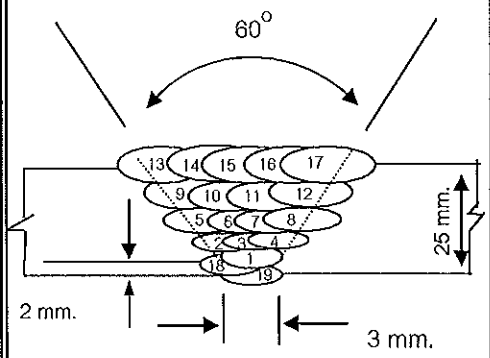
PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINERING CONSULTANT
69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Client :	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-009
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : RNC-WPS-002	Date : 6-Aug-12

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
-	MR.BOONTHAM J.	SMAW	2G	AWS D1.1	JIS G3101 Gr.SS400	25x290x380	A 5.1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
11	SMAW	E 7016	3.2	DCEP	165	26	14.53	200	17.72	-	-	-	-
12	SMAW	E 7016	3.2	DCEP	165	26	10.58	209	24.33	-	-	-	-
13	SMAW	E 7016	3.2	DCEP	166	26	13.95	212	18.57	-	-	-	-
14	SMAW	E 7016	3.2	DCEP	165	26	16.51	217	15.59	-	-	-	-
15	SMAW	E 7016	3.2	DCEP	167	26	14.83	242	17.57	-	-	-	-
16	SMAW	E 7016	3.2	DCEP	165	26	15.35	245	16.77	-	-	-	-
17	SMAW	E 7016	3.2	DCEP	164	26	15.63	246	16.37	-	-	-	-
18	SMAW	E 7016	3.2	DCEP	165	26	11.01	219	23.38	-	-	-	-
19	SMAW	E 7016	3.2	DCEP	164	26	6.87	181	37.24	-	-	-	-

PAE Inspection By	Witness By	Approved By
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Wichit K	NAME : Mr. Chatchawan S	NAME : Mr. Chatchawan S
DATE : 7-Aug-2012	DATE : 7/08/12	DATE : 7/08/12

PAE TECHNICAL SERVICE CO.,LTD.

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

TEL : 02) 322-0222 , FAX : 02) 721-2577 , RAYONG Tel : 038)607402-3, Fax : 038)607402, SRIRACHA : 038)330220, 330229

REPORT No.	PQR-014	RADIOGRAPHIC EXAMINATION REPORT	INSPECTION & TEST RECORDS
DWG.No.	PQR		
DATE :	6-Aug-12		

CLIENT :	Renew Coils Engineering & Supply Co.,Ltd.	PROJECT NAME :	PQR

PROCEDURE No. :	PAE-RT-001 (Rev.0)	EXPOSURE TIME		min. sec.	SOURCE TO FILM DISTANCE	17" Inch. / - mm.	
RADIATION SOURCE	X-ray (-) MAX - kVG. / γ - ray (Ir-192)	DENSITY RANGE		2.0 - 3.5	TYPE OF PENETRAMETER	ASTM-1B	
		SENSITIVITY		2%			
SOURCE RANGE	21.10 Ci.	FILM	TYPE	KODAK "MX"	FILM PROCESSING	AUTO	-
SOURCE SIZE	3 x 3 mm.		SIZE	3.5" x 17"		MANUAL	✓
TUBE VOLTAGE	- kVP.	INTENSIFYING SCREEN	FRONT	0.005 Inch.	MATERIAL	SS 400	
TUBE CURRENT	- mA		BACK	0.005 Inch.	THICKNESS	25 mm.	

APPLICABLE STANDARD : ASME Section V Article 2 — ACCEPTANCE CRITERIA : ASME-IX

RADIOGRAPHIC TECHNIQUE : ☑ SWSI ☐ DWSI ☐ DWDI

NAME / WQT	Position No.	Pipe/Plate Size Dia	Pipe/Plate Size Thk(mm)	Welder No.	Film No.	Interpretation	Results	Remarks
SMAW / WPS No.RNC-PQR-002								
MR.BOONTHAM J.	2G	-	25	-	A-B	NAD	ACC	

Total 3.5"x8.5" [-] Films

3.5"x17" [1] Films

ABBREVIATIONS :

ACC : Accepted
REJ : Rejected
CP : Cluster Porosity
SF : Surface Flaw
EUC : External Udercut
WH : Worm Hole
C : Crack
P : Porosity
UC : Undercutting
SL : Slag Inclusion
LP : Lack of Penetration
LF : Lack Of Fusion
IP : Incomplete Penetration
TI : Tungsten Inclusion
BT : Burn Through
CRP : Concave Root Penetration
ERP : Excess Root Penetration
NAD : No Apparent Defects
RS : Reshoot

PAE INSPECTED BY		REVIEWED BY		APPROVED BY	
SIGNATURE :		SIGNATURE :		SIGNATURE :	
NAME :	Mr.Sarawut P.	NAME :	Mr. Chatchawan S.	NAME :	
DATE :	7-Aug-12	DATE :	8/08/12	DATE :	

LPN
LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

LABORATORY ACCREDITATION TESTING 23852

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-12-08-080

Revision No. 0

Test Product	PQR No. RNC-PQR-002	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Test	Reduced Section Tension
Dimension	25 mm Thk.	Equipment's Capacity	200 TONS
Date Received	10 August 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	14 August 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	3179/12 TR1	N/A	24.93	20.31	506.328	N/A	N/A	N/A	219.276	433.071	N/A	OUT OF WELD
2	3179/12 TR2	N/A	24.79	20.07	497.535	N/A	N/A	N/A	216.727	435.602	N/A	OUT OF WELD

The unit of N/mm^2 is equivalent to MPa

Additional details : WPS No. RNC-WPS-002

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ...11.../...8.../...12...

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ...../...../.....

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited
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LPN PLATE MILL

23857

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1

Report No. BD-12-08-019

Revision No. 0

Test Product	PQR No. RNC-PQR-002	Material Characterization	Plate
Project Name	N/A	Test Method	AWS D1.1/D1.1 M:2010
Project No.	N/A	Standard of Test	AWS D1.1/D1.1 M:2010
Material Specification	SS400 / A36	Type of Jig	Guided-Bend Roller Jig
Dimension	25 mm Thk.	Diameter of Mandrel	38.1 mm
Date Received	10 August 2012	Bend Angle	180 Degree.
Date Tested	14 August 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	3179/12 BS1	Transverse side bend	9.92x23.75x161.86 mm	Open Discontinuity	WELD ZONE	0.58 mm
2	3179/12 BS2	Transverse side bend	9.97x23.79x161.88 mm	No Open Discontinuity	N/A	-
3	3179/12 BS3	Transverse side bend	9.87x23.84x162.14 mm	No Open Discontinuity	N/A	-
4	3179/12 BS4	Transverse side bend	9.91x23.83x162.08 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-002

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ...14.../...8.../...12...

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ...15/8/12...

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Testing Laboratory LPN Plate Mill Public Company Limited

12746

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Report No. HN-12-08-032

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

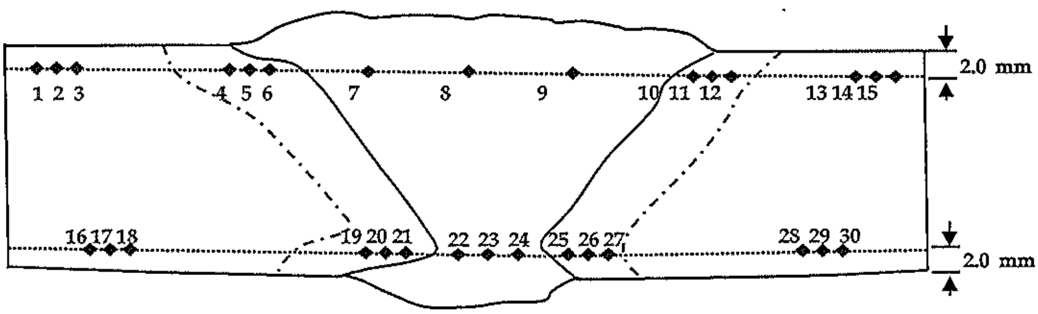
Revision No. 0

Test Product	PQR No. RNC-PQR-002
Test No.	3179/12 HN
Project Name	N/A
Project No.	N/A
Material Specification	JIS G3101 SS400
Dimension	25 mm Thk.
Welding Process/Position	SMAW/2G
Type of Joint	Butt joint
Date Received	10 August 2012
Date Tested	14 August 2012

Material Characterization	Plate
Test Method	ASTM E92-03
Standard of Test	AWS D1.1/D1.1M : 2010
Type of test	Vicker Hardness Test
Equipment	Mitutoyo, HV-115
Indentor Size	Diamond, pyramid 136 deg.
Test Load	HV 10 Kgf.
Total Force Dwell Time	15 s
Cal. Block	No. 62137 No. A62072
Ambient Temperature	25.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	140.1	16	BASE	142.9	******************************		
2	BASE	140.9	17	BASE	141.0			
3	BASE	141.6	18	BASE	142.3			
4	HAZ	160.4	19	HAZ	154.3			
5	HAZ	163.7	20	HAZ	160.2			
6	HAZ	170.8	21	HAZ	171.8			
7	WELD	171.6	22	WELD	170.0			
8	WELD	169.9	23	WELD	165.1			
9	WELD	164.9	24	WELD	175.3			
10	HAZ	164.8	25	HAZ	167.4			
11	HAZ	161.4	26	HAZ	157.3			
12	HAZ	161.2	27	HAZ	153.1			
13	BASE	140.0	28	BASE	141.4			
14	BASE	141.3	29	BASE	141.3			
15	BASE	142.0	30	BASE	141.7			

Additional details : Filler Metal : E7016(SMAW)

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurancer

Date : 15/08/12

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : ....

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Macrostructure Analysis Report

Report No. MA-12-08-023

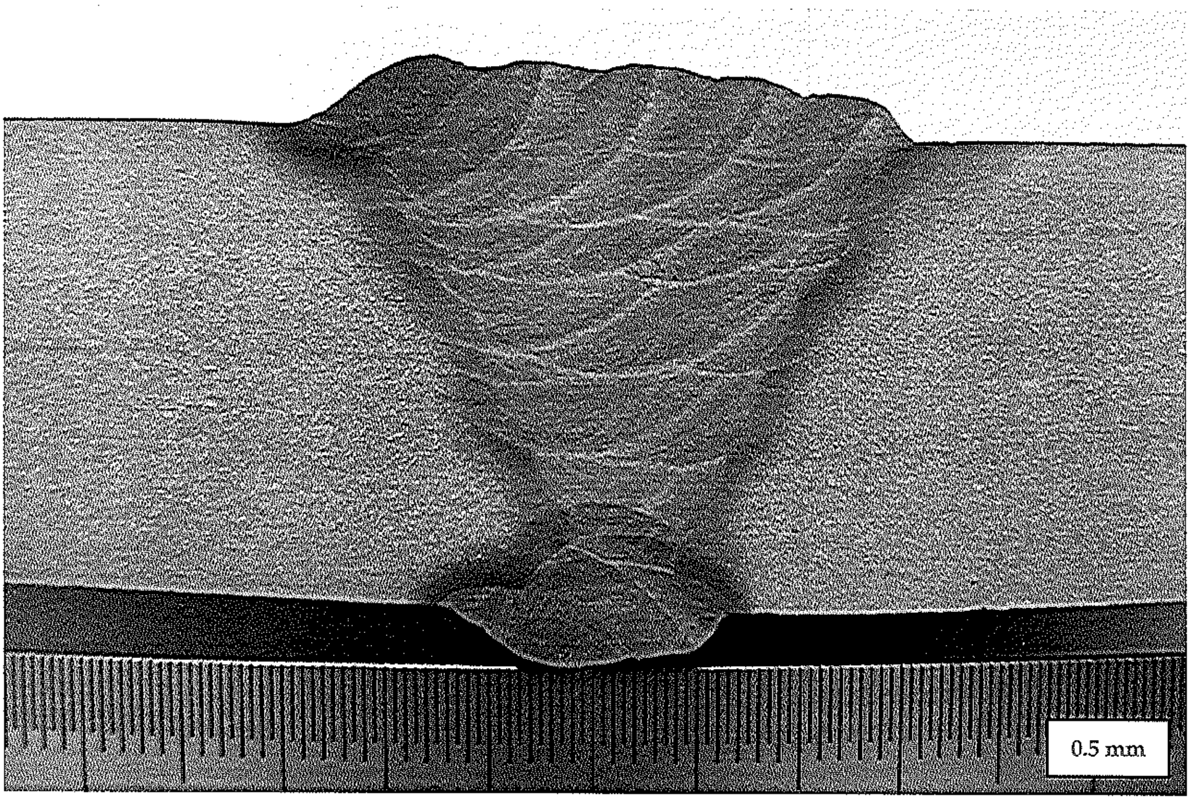
Revision No. 0

Customer : RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	PQR No. RNC-PQR-002
Test No.	3179/12 MA
Project Name	N/A
Project No.	N/A
Material Specification	JIS G3103 SS400
Dimension	25 mm Thk.
Welding Process/Position	SMAW/2G
Type of Joint	Butt joint
Date Received	10 August 2012

Material Characterization	Plate
Test Method	ASME E340-00
Standard of Test	AWS D1.1/D1.1M : 2010
Etchant Reagent	75ml H_2O+25ml HNO_3(65%)
Temperature	28 °C
Time of Etching	10 minutes
Surface Preparation	Polishing with sand paper 240, 400, 600, 800, 1000
Date Tested	14 August 2012



Additional details : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.
Filler Metal : E7016(SMAW)

RENEW COILS ENGINEERING & SUPPLY CO., LTD. RNC QC/QA

Reported by : ....
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurancer
Date : ..15../..08../..12..

Approved by : ....
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : ..15../..08../..12..

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LABORATORY ACCREDITATION 23862
TESTING No 0136

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-08-085

Revision No. 0

Test Product	PQR No. RNC-PQR-002
Project Name	N/A
Project No.	N/A
Material Specification	SS400 / A36
Dimension	25 mm Thk.
Date Received	10 August 2012
Date Tested	15 August 2012

Material Characterization	Plate
Test Method	ASTM A370-10 and ASTM E23M-07a[ε1]
Standard of Test	AWS D1.1/D1.1 M:2010
Type of Test	Charpy V-Notch Impact
Equipment / Serial No.	Tinius Olsen / 221811
Equipment 's Capacity	542 Joules
Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	3179/12-1	-20°C	10.017	10.018	54.98	192.42	177.33	Weld Metal	-
2	3179/12-2	-20°C	10.018	10.013	54.97	194.05			
3	3179/12-3	-20°C	10.017	10.012	54.96	145.52			
4	3179/12-1	-20°C	10.014	10.011	54.86	195.81	210.87	Fusion Line+1mm	-
5	3179/12-2	-20°C	10.018	10.014	54.86	262.31			
6	3179/12-3	-20°C	10.016	10.010	54.89	174.50			
7	3179/12-1	-20°C	10.010	10.017	54.97	9.99	10.24	Fusion Line+5mm	-
8	3179/12-2	-20°C	10.014	10.016	54.98	5.27			
9	3179/12-3	-20°C	10.013	10.013	54.99	15.46			

Additional details : WPS No. RNC-WPS-002

Welding Process : SMAW

Filler Metal/Electrode Spec't : E7016

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD. QC/QA

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date ...16/08/12...

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date ..../..../....

LPN PLATE MILL

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20/11/54

KOBELCO

INSPECTION CERTIFICATE

PURCHASER

Date of Issue 11/Aug/2011
Certificate No. TF11-035
Date of Inspection 16/Aug/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
LB-52	3.2 x 350	12110	AWS A5.1 E7016

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Ni	Cr	Mo	V	*1*
Weld Metal	0.07	0.52	0.93	0.009	0.003	< 0.01	0.03	< 0.01	0.009	0.97

Tensile Test of Weld Metal			Impact Test of Weld Metal			
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy		Tested size for Mechanical test: 4.0x400 mm Tested MFG. No. of Mechanical test: 12009
465 MPa	554 MPa	33 %	-30 °C	Avg.	246 J	

Welding Conditions				Postweld Heat Treatment	REMARKS
Type of Current	AC			--- °C x --- Hr.	*1* : Mn+Ni+Cr+Mo+V
Welding Current	120 A				
Welding Voltage	24 V				

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

THAI-KOBE WELDING CO., LTD.

CHIEF INSPECTOR

Certificate No. : 12-05-202

Reference : Order Sheet No. 524/26161

INSPECTION CERTIFICATE

SAHAVIRIYA PLATE MILL PCL

Office 28/1 Prapawit Bldg., 4th Fl., Surasak Rd., Silom, Bangrak, Bangkok, Thailand 10500 Tel.: (662)630-0257-66 Fax: (662) 630-0267

Factory 160 Moo 14, Sukhumwit Rd., Bangpakong, Chachoengsao, Thailand 24130 Tel.: (038)832-056-60 Fax: (038)531-635

Customer Name :

Commodity : HOT ROLLED STEEL PLATES

Specification : REFER TO BS 4360-1990 GRADE 43A

ISSUE DATE : 24/05/2012

Dimension (mm) Thick	Width	Length	Heat No.	Unit Weight (MT / Pc)	Number of Piece (s)	Weight MT (s)	Tensile Test: Tensile Strength (N/mm^2)	Yield Strength (N/mm^2)	Elongation (%) L_o = 200 mm	Bend (180)	Impact Test: Test Temp. C	Absorb Energy (Joule) Each KV_2	Avg KV_2
25.00	1,219	2,438	S2-06636	0.583	28	16.324	418	251	30	PASS	-		-
25.00	1,219	2,438	S2-06637	0.583	6	3.498	426	258	28	PASS	-		-
25.00	1,219	2,438	S2-06644	0.583	18	10.494	420	270	29	PASS			
25.00	1,219	2,438	S2-06645	0.583	38	22.154	415	264	28	PASS			
TOTAL					90	52.470							

Heat No.	C (x 100)	Si (x 100)	Mn (x 100)	P (x 1000)	S (x 1000)	Al (x 1000)	Nb (x 1000)	V (x 1000)	N (x 1000)	Mo (x 100)	Cr (x 100)	Ni (x 100)	Cu (x 100)	CEQ (x 100)
S2-06636	16	23	40	12	6	15	1	0		0	5	2	3	25
S2-06637	16	26	46	15	16	20	1	0		0	[illegible]	2	5	26
S2-06644	15	23	42	10	6	19	1	0		0	[illegible]	1	3	24
S2-06645	14	23	45	14	8	18	1	0		0	[illegible]	2	3	24

LABORATORY ENGINEER

WE HEREBY CERTIFY THAT MATERIAL DESCRIBED HEREIN HAS BEEN SATISFACTORY TESTED

AUTHORIZED